

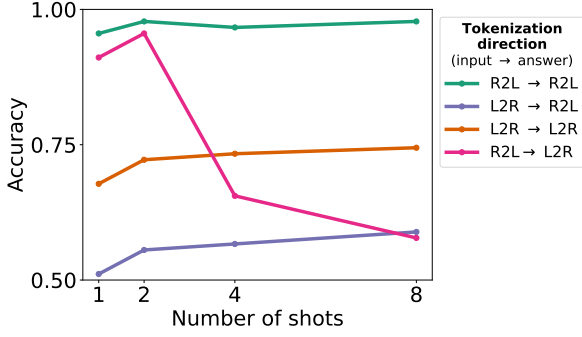


Figure 11. Few-shot accuracy when models receive a problem in one tokenization format, and answer it in another. The distinction between this and Figure 10 is that models *do not* repeat the problem in this case. We note that when giving a model a problem in R2L tokenization and prompting it to answer in L2R tokenization, the model actually gets *worse* with more shots, since for fewer shots, the model ends up ignoring the few-shot prompt and answers in its preferred R2L tokenization. Specifically, adherence to the prompted formatting for R2L→L2R increases from just 13.3% with 1 shot to 98.9% with 8 shots.

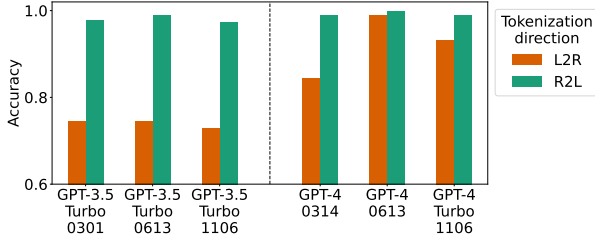


Figure 12. 8-shot performance of various OpenAI models on the same addition problems as Figure 4. The newer version of GPT-3.5 appears to perform equally poorly. For GPT-4, we see a large tokenization dependent effect in the March model, which becomes weaker (but still present) in the June model. The GPT-4 turbo model shows a slight regression in overall performance with the tokenization-dependent effect becoming stronger again.

As shown in Figure 12, we find that generally, yes: tokenization-dependent effects persist. We consider five “held-out” OpenAI models, which allow us to consider how tokenization-dependent effects shift when models are updated (gpt-3.5-turbo-0613, gpt-3.5-turbo-1106) or scaled up (gpt-4-0314, gpt-4-0613) and then scaled back down (gpt-4-1106-preview, which is a “turbo” model¹¹). Later versions of GPT-3.5 exhibit as strong an effect due to tokenization direction. The effect is mitigated slightly in GPT-4’s March version, and mitigated strongly in GPT-4’s most recent version.¹² Specifically, GPT-4 models appear

¹¹We assume this is a smaller, maybe distilled, version of GPT-4.

¹²We find it interesting that the March to June update to GPT-4 improved performance, but the corresponding update to GPT-3.5 did not – without knowing what these updates entail, however, it’s hard to draw conclusions as to why this may be the case.

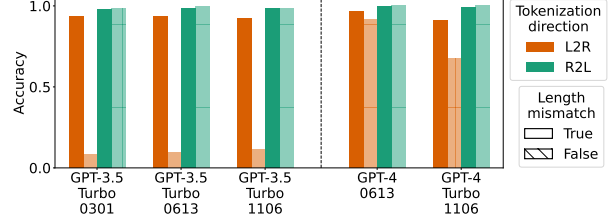


Figure 13. 8-shot performance of various OpenAI models on answer length controlled problems (see Section 4.1), separated by whether the answer length is the same as one of the addends. We see the effect from Section 4.1 reproduces strongly in the newer version of GPT-3.5. The effect is still present in GPT-4,¹⁴ but not as strongly. Interestingly, the effect is stronger in the latest GPT-4 Turbo model as compared to GPT-4.

to be better at performing arithmetic across the board (for both tokenization directions). Interestingly, in the most recent GPT-4 Turbo model, the effect of tokenization becomes stronger again. Furthermore, Figure 13 shows that the digit length mismatch between answer and addends is again the main reason for the performance drop when using L2R tokenization, in both GPT-3.5 and GPT-4 models. We believe that the increased scale of training GPT-4 (likely in both parameter count and data seen) allows it to better override the tokenization-induced inductive bias that leads GPT-3.5 models to perform worse (analogous to scale helping mitigate tokenization-induced spelling difficulties (Liu et al., 2023)). The resurgence of tokenization-dependent effects in the newest GPT-4 Turbo model (which is presumably smaller than GPT-4) supports this hypothesis.

7. Related work

Tokenization methods The two leading tokenization methods are Unigram (Kudo, 2018) and BPE (Sennrich et al., 2016). While older work in NLP show the benefits of Unigram over BPE (Bostrom & Durrett, 2020), BPE remains the most commonly used tokenization method by modern LLM practitioners. Within BPE, different models often make different hard-coded choices, such as removing long tokens of consecutive whitespace (Touvron et al., 2023a) or enforcing single-digit tokenization of numbers (Chowdhery et al., 2023). Our work demonstrates tokenization-dependent effects from one such choice, the use of 1-, 2-, and 3- digit tokens by OpenAI models (OpenAI et al., 2023). One way around such issues could be tokenizer-free methods (e.g., MEGABYTE (Yu et al., 2023), which uses patch-based schemes and doesn’t assume fixed tokens), but we suspect these schemes will also carry their own inductive biases. Golkar et al. (2023) also introduce a continuous number encoding scheme meant to circumvent tokenization artifacts, but their approach is limited to cases where model outputs are purely numerical, and not interleaved with text.

Tokenization artifacts in LLMs A growing set of results has emerged around various tokenization-related artifacts in LLMs. Similar to scratchpad prompting (Nye et al., 2021), Wei (2023) found that separating letters into individual tokens can help in sorting words by the second letter. Other work (Shin et al., 2020) has focused on specific tokens that can negatively affect model performance. Rumbelow & mwatkins (2023) found many tokens which were artifacts of the data used to pre-train the tokenizer, but presumably weren’t present in the model’s training data, leading to highly unpredictable (and often comical) completions. Sun et al. (2023) find artifacts due to mismatches in tokenization in extractive Q&A tasks, which may have connection to some of our experiments in Section 5.2. Lundberg (2023) propose *token healing* to avoid many tokenization-related issues by removing the last few tokens from a prompt and allowing the model to complete them; this approach has connections to work on asking models to *rephrase-and-respond* (Deng et al., 2023) and our experiments on prompting the model to repeat with its preferred tokenization direction (Section 5). Our work builds on these past scattered artifacts and provides a systematic analysis of tokenization-direction-dependent effects on numerical reasoning in frontier LLMs.

Arithmetic tasks as a testbed for numerical reasoning in LLMs With the increased interest in measuring frontier models on math reasoning (Saxton et al., 2019; Cobbe et al., 2021; Lewkowycz et al., 2022; Hendrycks et al., 2021; Paster, 2023), an accompanying body of work studies language models in more controlled settings, such as arithmetic. Razeghi et al. (2022) use arithmetic tasks to show that pre-training term frequencies¹⁵ can affect numerical reasoning in GPT-J models (Wang & Komatsuzaki, 2021) trained on the Pile dataset (Gao et al., 2020). Similarly, McCoy et al. (2023) showed that GPT-3.5 and GPT-4 are better at computing linear functions that are more common in training data (such as the Fahrenheit to Celsius conversion) than close alternatives. Other work (Nogueira et al., 2021; Muffo et al., 2022; Zhou et al., 2022; 2023) instead focuses on various modifications that can help models generalize to longer arithmetic tasks. Zhou et al. (2023) and Lee et al. (2023) both point out that having autoregressive models perform addition in reversed order yields a simpler algorithm to learn and results in better performance, which is complementary to our emphasis on the importance of “reversed” (i.e. right-to-left) tokenization alignment. Zhou et al. (2022) also conduct preliminary error analyses of model mistakes, though their algorithmic prompts force models to split tokens into digits (similar to Nye et al. (2021)). Our work broadly lies in this

¹⁵We also considered frequency effects by utilizing BPE merge ranks (given that we do not have access to the pre-training data of GPT-3.5 and GPT-4 models) as an approximate for frequency. We didn’t find a strong effect, expanded results are provided in Appendix C.

category of work using arithmetic tasks to study numerical reasoning; we chose to focus on tokenization-dependent effects, and found surprisingly consistent, stereotyped error patterns (Section 4), adding to this rich body of literature.

8. Discussion

In this work, we analyze tokenization-dependent effects on numerical reasoning in GPT-3.5 and GPT-4. We found that the hard-coded choice of 1-, 2- and 3- digit number tokens, tokenized left-to-right, gives rise to stereotyped error patterns when compared to right-to-left tokenization. We proposed a mitigation, where the model is asked to repeat the answer in its preferred tokenization format. Finally, we showed that the effect is stronger in smaller models (such as GPT-3.5), emphasizing the significance of tokenization-dependent inductive biases in an era where many practitioners are focusing on packing capabilities into smaller models through overtraining (De Vries, 2023; Touvron et al., 2023a) and distillation (Li et al., 2023; Gemini Team et al., 2023). Overall, we believe this evidence strongly suggests inductive biases from tokenization can significantly influence model performance on numerical reasoning tasks.

Modern frontier LLMs mostly use single-digit tokens (Table 1), with GPT-3.5 and GPT-4 being a key exception in their use of up-to-3-digit tokens. We hypothesize that the latter choice may have been made to achieve a better *compression rate*: models “see” more numerical data for the same number of training tokens.¹⁶ Furthermore, this choice could have benefits for *length generalization* (Anil et al., 2022), as we allude to in Section 4.4. However, we’ve also demonstrated how the misalignment between inputs and outputs when using L2R tokenization (Section 4.1) can lead to large drops in accuracy, especially on smaller models (GPT-3.5, GPT-4 Turbo). Such misalignment would not be an issue when using single-digit tokens.

To make progress on which number tokenization choices are best to use (e.g., the single-digit tokens of LLaMa and PaLM, or the up-to-3 digit tokens of GPT-3.5 and GPT-4), the “gold experiment” would be to train the same model architecture on the same dataset, but with varying number tokenization strategies. Beyond the expense of this experiment (making it intractable in academic settings), a key question also becomes how to “compute”-control. The better compression ratio of up-to-3 digit tokens means a token-controlled experiment would result in some models “seeing” more data. We hope our work leads model practitioners to consider such ablations, with proper controls.

Beyond applicability to model practitioners, our work also

¹⁶3-digit number tokens also reduce inference-time compute when models use numbers in their output, which could be an important consideration when serving models at scale.

provides an interesting set of tokenization-dependent phenomenon for interpretability researchers to explore. Prior work (Stolfo et al., 2023) has used techniques such as *path patching* to identify sub-circuits in LLMs that perform arithmetic tasks, but restricted to single token operands. Building off our results, it would be interesting to elucidate the mechanisms behind systematic error patterns, especially in the case of multi-token operands. The robustness of the “digit 4” error on GPT-3.5 points to some systematic mechanism, which could shed light on underlying algorithms that emerge to perform arithmetic tasks.

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